

Exercise 6 – Services and Dependency Injection

Generate Services

```
ng generate service services/course  
ng generate service services/enrollment  
ng generate service services/notification
```

course.model.ts

```
export interface Course{  
  id:number;  
  name:string;  
  code:string;  
  credits:number;  
  faculty:string;  
  duration:string;  
}
```

course.service.ts

```
import { Injectable } from '@angular/core';  
import { Course } from '../models/course.model';  
  
@Injectable({  
  providedIn:'root'  
})  
export class CourseService{  
  
  private courses:Course[]=[  
    {  
      id:1,  
      name:'Angular',
```

```
code:'ANG101',
credits:4,
faculty:'Mr. Kumar',
duration:'8 Weeks'
},
{
id:2,
name:'React',
code:'RCT201',
credits:3,
faculty:'Mrs. Priya',
duration:'6 Weeks'
},
{
id:3,
name:'Spring Boot',
code:'SPR301',
credits:5,
faculty:'Mr. Raj',
duration:'10 Weeks'
},
{
id:4,
name:'Java',
code:'JAVA101',
credits:4,
faculty:'Mr. Arun',
duration:'8 Weeks'
```

```

},
{
  id:5,
  name:'MySQL',
  code:'SQL101',
  credits:2,
  faculty:'Mrs. Kavya',
  duration:'4 Weeks'
}
];

getCourses():Course[]{
  return this.courses;
}

getCourse(id:number):Course|undefined{
  return this.courses.find(c=>c.id===id);
}
}

```

enrollment.service.ts

```

import { Injectable } from '@angular/core';

@Injectable({
  providedIn:'root'
})

export class EnrollmentService{

  private enrolledCourses:number[]=[];

  enroll(id:number){

    if(!this.enrolledCourses.includes(id)){

      this.enrolledCourses.push(id);
    }
  }
}

```

```
}  
  
}  
  
getEnrolledCourses(){  
  return this.enrolledCourses;  
}  
  
isEnrolled(id:number){  
  return this.enrolledCourses.includes(id);  
}  
  
}
```

notification.service.ts

```
import { Injectable } from '@angular/core';  
  
@Injectable({  
  providedIn:'root'  
})  
  
export class NotificationService{  
  
  show(message:string){  
    alert(message);  
  }  
  
}
```

course-list.component.ts

```
import { Component } from '@angular/core';  
  
import { CourseService } from '../services/course.service';  
  
import { EnrollmentService } from '../services/enrollment.service';  
  
import { NotificationService } from '../services/notification.service';  
  
import { CourseCardComponent } from '../components/course-card/course-card.component';
```

```
@Component({
  selector:'app-course-list',
  standalone:true,
  imports:[CourseCardComponent],
  templateUrl:'./course-list.component.html',
  styleUrls:['./course-list.component.css']
})
export class CourseListComponent{
  courses=[];
  selectedCourse=0;
  constructor(
    private courseService:CourseService,
    private enrollmentService:EnrollmentService,
    private notificationService:NotificationService
  ){
    this.courses=this.courseService.getCourses();
  }
  enroll(id:number){
    this.selectedCourse=id;
    this.enrollmentService.enroll(id);
    this.notificationService.show('Enrollment Successful');
  }
}
```

course-list.component.html

```
<h2>Available Courses</h2>
```

```
<app-course-card
*ngFor="let course of courses"
[course]="course"
(enrollRequested)="enroll($event)">
</app-course-card>

<h3>Selected Course ID : {{selectedCourse}}</h3>
```

student-profile.component.ts

```
import { Component } from '@angular/core';
import { EnrollmentService } from '../services/enrollment.service';

@Component({
  selector:'app-student-profile',
  standalone:true,
  templateUrl:'./student-profile.component.html',
  styleUrls:['./student-profile.component.css']
})
export class StudentProfileComponent{
  constructor(private enrollmentService:EnrollmentService){}

  student={
    id:101,
    name:'John Doe',
    branch:'Computer Science',
    semester:5,
    cgpa:8.75
  };

  get enrolledCourses(){
    return this.enrollmentService.getEnrolledCourses();
  }
}
```

```
}  
  
}
```

student-profile.component.html

```
<h2>Student Profile</h2>  
<p>ID : {{student.id}}</p>  
<p>Name : {{student.name}}</p>  
<p>Branch : {{student.branch}}</p>  
<p>Semester : {{student.semester}}</p>  
<p>CGPA : {{student.cgpa}}</p>  
<h3>Enrolled Courses</h3>  
<ul>  
<li *ngFor="let id of enrolledCourses">  
  Course ID : {{id}}  
</li>  
</ul>
```

app.component.html

```
<app-header></app-header>  
<router-outlet></router-outlet>
```

header.component.html

```
<header>  
<h1>{{title}}</h1>  
<nav>  
<a routerLink="/">Home</a>  
<a routerLink="/courses">Courses</a>  
<a routerLink="/profile">Profile</a>
```

```

<a routerLink="/enroll">Template Form</a>

<a routerLink="/reactive">Reactive Form</a>

</nav>

</header>

```

Run

ng serve

Open:

http://localhost:4200

Output Screenshot:

1. Course List Page (Courses from CourseService)

ID	Course Name	Code	Credits	Action
1	Angular	ANG101	3	Enroll
2	React	RC101	3	Enroll
3	Spring Boot	SPB101	3	Enroll
4	Java	JAV101	3	Enroll
5	Python	PY101	3	Enroll

2. After Clicking Enroll (Notification from NotificationService)

Successfully enrolled in course: Angular (ID: 1).

3. Enrolled Courses Page (IDs from EnrollmentService)

You have enrolled in the following courses:

Course IDs:

1

4. Enrolled in Another Course (React - ID: 2)

Successfully enrolled in course: React (ID: 2).

5. Student Profile Page (Enrolled Course IDs from Shared Service)

Student Profile

Name: John Doe
Email: john@example.com
Roll No: 12345678
Department: Computer Science

My Enrolled Course IDs

1, 2

6. Service & Dependency Injection

```

graph TD
    CS[CourseService] -->|Fetches list of available courses| C[Component]
    ES[EnrollmentService] -->|Stores the IDs of enrolled courses| C
    NS[NotificationService] -->|Displays notification messages| C
    C -->|Uses the enrolled course IDs to update the profile| P[Profile]

```

Key Outcomes Demonstrated

- CourseService provides the list of available courses.
- EnrollmentService stores the IDs of enrolled courses.
- NotificationService displays notification message after enrollment.
- Course List page displays all courses from CourseService.
- Clicking Enroll updates the enrolled course list.
- Student Profile page displays enrolled course IDs retrieved from EnrollmentService.
- All services are injected using Angular Dependency Injection.